4. Direct hunting and persecution of species leading to 'selective mass extinction.

5. Ecological substitution by other species of large carnivorous animals which compete for the same food resources.

6. Climatic change accelerates the competition between large mammals for shelter and food.

7. Extinction of weak species during the course of competition with more powerful and stronger species.

8. Man-induced environmental changes also cause species extinctions.

Between 1600 and 1900 it is estimated that one species went extinct every four years. In modern times, the rate is soaring. The graph below (Figure 7.12.) shows how the rate of extinction of species has increased over the past 50 years. This could be attributed to the rapid increase in population during the same period of time.

According to IUCN the rate of extinction of mammals and birds had started much earlier by 1700 itself at a much faster rate as shown in the graph below (Figure 7.13).

7.6.2 Major Threats to Biodiversity

The following are some of the major threats to biodiversity:

a. Habitat destruction and degradation

b. Invasive alien species-these can destroy native species Example, lantana Camera plant in India.

c. Climate Change- Example, bleaching and loss of coral reefs due to global warming

d. Pollution of air, water and soil – Pollution can alter the growth and life of organisms in a great way.

e. Over exploitation of one resource – Over exploitation through Hunting or Poaching, Deforestation etc., can influence the life of all the interdependent species.

Despite rapid efforts in protecting terrestrial and marine habitats, world's diversity of species is still dwindling. Since the 1960's over 100,000 'protected areas' have been established. This represents 11,265,408 sq.km of land and 1,609,344 sq.km of ocean. Yet, terrestrial and marine species have declined over the same period. This suggests that the common conservation strategy of protecting areas of land and sea is inadequate.

7.7. Conservation of Biodiversity

Conservation of bio-diversity is the proper management of the biosphere by human beings in such a way that it gives maximum benefits for the present generation and also develops its potential to meet the needs of the future generations.

The three basic objectives of biodiversity conservation are :

(a) To maintain essential ecological processes and life supporting systems.

(b) To preserve the diversity of species.

(c) To make sustainable utilization of species and ecosystems.

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